

EDUCATION

The Chinese University of Hong Kong (CUHK)	Hong Kong SAR
<i>Doctor of Philosophy , Geophysics (Seismology)</i>	2026 – 2030
Kharazmi University, Faculty of Earth Sciences	Tehran, Iran
<i>Bachelor of Science, Geology</i>	2021 – 2025
- Chairperson, Geological Society of Kharazmi University (June 2024 – June 2025) - Vice President of Media, Geological Society of Kharazmi University (November 2023 – June 2024)	
University of Lille, International Academy Lille Hdf	Lille, France
<i>Water Resource Management Summer School (5 ECTS)</i>	Summer 2024
Presented a numerical groundwater modeling study on water abstraction impacts in the Weyib Watershed, Ethiopia. Under Supervision of the Prof. Arnaud Gauthier.	

TEACHING EXPERIENCE

CUHK	Hong Kong SAR
<i>Hydrogeology Teaching Assistant</i>	August 2026 – Present
(EESC3320), under supervision of Prof. Yen Joe Tan	
Space Research and Technology Institute (SRTI)	Primorsko, Bulgaria
<i>Lecturer</i>	August 2026 – September 2026
EPEC ML WG Summer School 2026, under supervision of Prof. Lachezar Filchev	
Iranian Young Scholars Club (IrYSC)	Tehran, Iran
<i>Earth Science Olympiad Teacher</i>	April 2021 – August 2026
Kharazmi University	Tehran, Iran
<i>Programming Teaching Assistant</i>	April 2025 – June 2025
Programming Course, under supervision of Prof. Morteza Asemani	
Kharazmi University	Tehran, Iran
<i>Geostatistics Teaching Assistant</i>	February 2025 – June 2025
Geostatistics Course, under supervision of Prof. Ehsan Pegah	
Young Scholars Club	Tehran, Iran
<i>Instructor</i>	July 2024
Taught "Applied Data Science in Geology" for the Iran National IESO¹ 2024 Team	
Kharazmi University	Tehran, Iran
<i>Instructor</i>	April 2024 – June 2024
Organized and instructed the Advanced Python and Machine Learning Workshop	
NODET²	Tehran, Iran
<i>Teacher</i>	May 2021 – September 2021
Taught ITEO³ Marketing at Allameh Helli (Branches 5 & 8) and Farzanegan High Schools	

RESEARCH EXPERIENCE

FEZtool	Tehran, Iran
<i>Co-Founder & Research and Development Specialist</i>	August 2023 – Present
Led Earth science research using advanced algorithms and geospatial analysis.	
Center for Complex Networks, SBU Partner	Tehran, Iran
<i>Research Assistant</i>	November 2025 – June 2026
Worked on fashion trend forecasting project under supervision of Prof. Reza jafari.	
International Astronomical Search Collaboration, NASA Partner	Remote
<i>Asteroid Search Team Member</i>	July 2025 – September 2025
Analyzed Pan-STARRS data, contributing to 3 preliminary asteroid discoveries.	
Kharazmi University	Tehran, Iran
<i>Research Assistant</i>	October 2024 – January 2025
Performed MASW⁴ to extract dispersion images under supervision of Prof. Mehdi Talkhablou.	

¹International Earth Science Olympiad²National Organization for Development of Exceptional Talents³Iranian Thinking & Entrepreneurship Olympiad

COMMUNITY SERVICE

Zamin Pouya Scientific Journal Vol. 31

Associate Editor & Reviewer

The 4th National Festival of Young Iranian Geologists

Education Committee Member

The 3rd National Festival of Young Iranian Geologists

Graphics Committee Member

Imam Khomeini International University

Editorial Board Member

Tehran, Iran

August 2025 – June 2026

Tehran, Iran

April 2025 – May 2025

Tehran, Iran

May 2024 – June 2024

Qazvin, Iran

September 2023 – June 2024

SELECTED PUBLICATIONS

Journal Articles

1. Talkhablou, M., **Farmahini Farahani, M.**, & Mansouri, S. S. (2025). A Python-Based Framework for Land Cover Classification in Engineering Geology: A Comparative Assessment of SVM, K-Means, and Spectral Indices. *Kharazmi Journal of Earth Sciences*.
2. **Farmahini Farahani, M.** (2024). Designing the green belt in the southwest of Zahedan. *Ovan Scientific Journal*
3. Saravani, M. J., Kashef, S., **Farmahini Farahani, M.**, Kashefi, M., & Zohreh, M. (2023). Investigating the accuracy of hybrid models with wavelet transform in the forecast of watershed runoff. *Journal of Water Management Modeling (JWMM)*

Conference Papers

1. **Farmahini Farahani, M.** (2025, August). Machine Learning-Enhanced Seismic Event Detection and Noise Reduction for Planetary Seismology: Insights from Mars and Lunar Data. In *Seismology Student Workshop (SSW 2025)*, GatherTown (Virtual)
2. Rezaei, K., & **Farmahini Farahani, M.** (2024, March). Visualization of magnetometric data with Python (case study: Segregation sedimentary units of Torud Playa). In *The 1st Conference on Applied Geophysics in Mines*, Tehran, Iran
3. Rezaei, K., & **Farmahini Farahani, M.** (2024, February). Comparing the performance of the k-means clustering method and spectral indices in characterizing land use and land cover using remote sensing with Python (case study: Landsat 8 multi-spectral images). In *2nd International Conference for Iranian Geography and Earth Sciences Students*, Tehran, Iran
4. **Farmahini Farahani, M.**, Rezaei, K., Karamollahi, S., & Moghadam, F. J. (2024, January). Application of PCA algorithm in remote sensing using Python programming language (case study: Landsat 9 multispectral images). In *8th Symposium of Sedimentological Society of Iran*, Hormozgan, Iran
5. Rezaei, K., & **Farmahini Farahani, M.** (2023, September). The importance of using Python programming language in Earth science. In *26th Conference of Geological Society of Iran*, Urmia, Iran

Books

1. **Farmahini Farahani, M.**, Najibifar, Y., & Akbarzadeh Afshari, A. (2025). *Earthquake Prediction Using Geophysical Data Analysis and Artificial Intelligence*. **Heritage Branch, Library and Archives Canada**
2. Rezaei, K., **Farmahini Farahani, M.**, Karimzadehasl, E. (Trans.). (2024). *Introduction to Python in Earth science data analysis: From descriptive statistics to machine learning*. **Tasir Publication**. (Original work published by Maurizio Petrelli)

PROJECTS

PlanetaryGeophysics | github.com/aradfarahani/PlanetaryGeophysics

2025 – Present

- Built an open-source Python platform for analyzing and visualizing planetary geophysical data.

FEZrs (Open Source) | pypi.org/project/fezrs

2023 – Present

- Advanced Python library for remote sensing, developed by FEZtool with over 100,000 downloads

Seismicity Analysis in Iran (2010–2025) | iris.edu/hq/workshops/2025/06

2025

- performed Iran seismicity analysis for the *Seismology Skill-Building Workshop 2025* final project.

⁴Multichannel Analysis of Surface Waves

- Interactive 3D visualization tool for subsurface resistivity using geoelectrics survey data

INVITED TALKS

1. **Walrus: A Cross-Domain Foundation Model for Continuum Dynamics:** *Applied ML Lab, Center for Complex Networks, Department of Physics, Shahid Beheshti University*; 13 December 2025
2. **Applied Artificial Intelligence in Earth Science:** *Faculty of Geosciences, Shahrood University of Technology & Faculty of Earth Sciences, Kharazmi University*; 11 November 2025
3. **Applications of Artificial Intelligence in Education and Earth Science Analysis:** *Educational Technology Expert Group, Karaj and Fardis County Department of Education, District 2*; 30 December 2024

LICENSES & CERTIFICATIONS

- Applied Geotechnical Seismology**, Kharazmi University 2026
- Applications of Non-Destructive Geophysical Methods in Engineering Geology Studies**, Kharazmi University
- Seismology Skill Building Workshop 2025**, EarthScope Consortium 2025
- Investigating practical issues in the exploration of Sulphide Deposits using the IP-RS method**, National Iranian Geophysical Society
- Remote Pilot Certification (Multirotor)**, Civil Aviation Authority of Islamic Republic of Iran 2023
- Unmanned Aerial Vehicle Surveying**, Maan Academy
- GIS, Mapping, and Spatial Analysis Specialization**, University of Toronto (Offered by Coursera)
- CS50 Certificate**, CS50 (Harvard University) 2021

HONORS & AWARDS

- CUHK's International Student PhD Fellowship - Awardee**, CUHK 2026
- 9th National Young Scientists Festival - Merit Award Winner**, Geology Category 2025
- Hydro 2024 Research-Oriented Summer School - Full Scholarship**, University of Lille 2024
- Research Visit Grant (2500 DKK) - Awardee**, University of Copenhagen 2022
- DrCT 2021 Global Computational Competition - Gold Medalist** 2021
- 1st national Thinking & Entrepreneurship Olympiad (ITEO) - Silver Medalist** 2020
- WMI 2015 World Mathematical Championships Selection - Silver Medalist** 2015

SKILLS & INTERESTS

Programming Languages: Python (ObsPy, pyGIMLi, Astropy, OpenCV, TensorFlow, PyGMT, Xarray, Scikit-learn, kepler.gl, Leafmap, Folium, Rasterio, GeoPandas), Julia, Bash (Shell scripting), SQL (MySQL), GMT, GDAL; Familiar with R

Markup Languages: Markdown, L^AT_EX

Softwares: Astrometrica, ENVI, Seismic Analysis Code(SAC), SeisAn, QGIS, SeisImager, Blender, ilastik, Google Earth Pro, Refrapy, SAS.Planet, draw.io, Groundwater Vistas

Field Techniques: Drone Surveying, Geoelectrical Methods (Induced Polarization–Resistivity Sounding, IP-RS), Seismic Refraction Data Acquisition

Languages: Farsi (Persian): *Native*; English: *Fluent (C1)*; German: *Elementary (A2)*; Mandarin: *Beginner (Duolingo Score: 10)*

Research Interests: Signal/Image Processing, Geophysics (Seismology), Remote Sensing, Machine Learning, Open-Source Software Development

PROFESSIONAL MEMBERSHIPS

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| National Iranian Geophysical Society | Student member, 2025 – Present |
| Space Generation Advisory Council | Member, 2025 – Present |
| Google Maps Platform Innovators | Member, 2025 – Present |
| Geological Society of Iran | Student member, 2021 – 2022 |